Sample space first

An experiment has sample space S of possible outcomes; an event is a subset of S. Classical $P(E) = |E|/|S|$ is valid only if the chosen outcomes are equally likely. Two fair dice use 36 equally likely ordered pairs, not 21 equally likely unordered pairs.

INTUITION

Probability is a normalized count

For equally likely finite outcomes, probability is a fraction of the whole sample space. For unequal outcomes, weight them by their probabilities instead of counting them uniformly.

Axioms: $P(S) = 1$, $P(E) \geq 0$, and disjoint events add. Hence $P(\varnothing) = 0$, $0 \leq P(E) \leq 1$, and if $E \subseteq F$, then $P(E) \leq P(F)$.

Complement principle

$$P(E^c) = 1 - P(E).$$

For "at least one", the complement "none" often has fewer cases. For n independent trials, success chance p:

$$P(\text{at least one success}) = 1 - (1-p)^n.$$

WORKED EXAMPLE

Two dice

Sum 7 has six ordered outcomes among 36: probability $1/6$. Writing the ordered space avoids weighting errors: unordered pair $\{1,6\}$ has twice the probability of $\{1,1\}$.

Equally likely versus independent

Equally likely is a property of outcomes in a model; independence is a relation between events. Independent events satisfy $P(A \cap B) = P(A)P(B)$. Neither notion implies the other.

COMMON TRAP

Sample-space checklist

State what is selected, whether order matters, replacement rules, and why outcomes have equal weight. "Favourable over total" without those assumptions is not a probability argument.

Overlap correction

For events A, B, disjointly split their union to obtain

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

For disjoint events the intersection is zero; for mutually exclusive events with positive probabilities, they cannot be independent.

WORKED EXAMPLE

Heart or king

In a 52-card deck, 13 hearts and 4 kings overlap at the king of hearts: $P(H \cup K) = (13 + 4 - 1)/52 = 4/13$.

Counting under a probability model

When all selections are equally likely, count using permutations if order matters, combinations if not, then divide. For a committee of size r from N, total outcomes $\binom{N}{r}$. "At least one" often uses a complement count.

WORKED EXAMPLE

Committee has a woman

Choose 3 from 5 men and 4 women. Probability at least one woman is $1 - \binom{5}{3}/\binom{9}{3} = 37/42$. The event and total use the same uniformly selected committee model.

Conditional probability shrinks the space

For $P(B) > 0$:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

Multiplication rule: $P(A \cap B) = P(A \mid B)P(B)$. Order matters: $P(A \mid B)$ generally differs from $P(B \mid A)$.

WORKED EXAMPLE

Even die, then six

Given an even result, the reduced space is $\{2,4,6\}$. Exactly one outcome is 6, so $P(6 \mid \text{even}) = 1/3$.

COMMON TRAP

Without replacement

The second draw's probability changes after the first. Use conditional probability; do not multiply the original marginal probabilities as if independent.

Independence is factorisation

Events A, B are independent iff $P(A \cap B) = P(A)P(B)$; equivalently, if $P(B) > 0$, $P(A \mid B) = P(A)$. Pairwise independence of three events does not guarantee mutual independence.

COMMON TRAP

Exclusive is not independent

If A, B are disjoint and both have positive probability, $P(A \cap B) = 0$ but $P(A)P(B) > 0$, so they are dependent.

Partition into causes

If $B_1, \ldots, B_n$ are disjoint and cover S, then

$$P(A) = \sum_i P(A \mid B_i)P(B_i).$$

The partition must be exhaustive and its branches disjoint.

Bayes reverses the tree

$$P(B_i \mid A) = \frac{P(A \mid B_i)P(B_i)}{\sum_j P(A \mid B_j)P(B_j)}.$$

Numerator = prior × likelihood; denominator = total probability of observed evidence.

WORKED EXAMPLE

Factory defect

Machines A,B,C make 50%,30%,20%, with defect rates 1%,2%,3%. Overall defect chance is $0.5(0.01) + 0.3(0.02) + 0.2(0.03) = 0.017$. Given defective, source A probability is $0.005/0.017 = 5/17$.

COMMON TRAP

Base-rate trap

Sensitivity $P(+ \mid D)$ is not the posterior $P(D \mid +)$. The latter depends on prevalence and false-positive rate through Bayes' denominator.

A random variable turns outcomes into numbers

Discrete X has pmf $p(x) = P(X = x)$, with $p(x) \geq 0$ and $\sum_x p(x) = 1$. For finite support:

$$E[X] = \sum_x x p(x), \qquad E[X^2] = \sum_x x^2 p(x)$$

$$\operatorname{Var}(X) = E[X^2] - (E[X])^2.$$

Standard deviation $\sigma = \sqrt{\operatorname{Var}(X)}$ has the original units; variance has squared units.

SHORT DERIVATION

Variance as spread about the mean

Expand $E[(X-\mu)^2]$ with $\mu = E[X]$: $E[X^2] - 2\mu E[X] + \mu^2 = E[X^2] - \mu^2$. Variance is nonnegative and is zero exactly when X is constant with probability 1.

Linear changes and indicators

$E[aX + b] = aE[X] + b$; $\operatorname{Var}(aX + b) = a^2 \operatorname{Var}(X)$. For indicator I_A, $E[I_A] = P(A)$ and $\operatorname{Var}(I_A) = p(1-p)$.

WORKED EXAMPLE

Three-point distribution

If $X = 1, 2, 3$ with probabilities $0.2, 0.3, 0.5$, then $E[X] = 2.3$, $E[X^2] = 5.9$, variance $= 5.9 - 2.3^2 = 0.61$, SD ≈ 0.781.

COMMON TRAP

Variance of a sum

In general $\operatorname{Var}(X+Y) = \operatorname{Var}X + \operatorname{Var}Y + 2\operatorname{Cov}(X,Y)$. The two-variance shortcut needs zero covariance (independence suffices); expectation linearity needs no independence.

Binomial: fixed independent trials

n independent Bernoulli trials, each success chance p, $q = 1 - p$. Number of successes X has

$$P(X = k) = \binom{n}{k} p^k q^{n-k}, \quad k = 0, 1, \ldots, n$$

$$E[X] = np, \qquad \operatorname{Var}(X) = npq.$$

The choose factor selects which k trials succeed; each such pattern has probability $p^k q^{n-k}$.

WORKED EXAMPLE

Exactly three successes

For 10 independent trials with $p = 0.2$, $P(X = 3) = \binom{10}{3}(0.2)^3(0.8)^7 \approx 0.201327$. "At least" can be summed or complemented.

COMMON TRAP

Model conditions

Fixed number of trials, two outcomes per trial, constant success probability, and independence. Without-replacement sampling is generally hypergeometric, not binomial.

Geometric waiting time

Let X count trials up to and including first success, $0 < p \le 1$. Then $P(X = k) = q^{k-1}p$ for $k \ge 1$, $E[X] = 1/p$, $\operatorname{Var}(X) = q/p^2$. If instead counting failures before success, support begins at 0; state the convention.

INTUITION

Why the mean is reciprocal

The tail probabilities are $P(X \ge k) = q^{k-1}$. Their sum is $E[X] = 1 + q + q^2 + \cdots = 1/(1-q) = 1/p$.

A geometric waiting time is memoryless: $P(X > m + n \mid X > m) = P(X > n)$ for nonnegative integers m, n.

Data and frequency notation

Observations x_i with frequencies f_i; total $N = \sum f_i$. For ungrouped data, frequencies may all equal 1. Arithmetic mean:

$$\bar{x} = \frac{\sum_i f_i x_i}{N}.$$

It is the balance point and minimizes the sum of squared deviations, not the sum of absolute deviations.

Median and mode

Sort the N observations. For odd N, median is position $(N+1)/2$; for even N, average of positions $N/2$ and $N/2 + 1$. Mode is the most frequent value; multiple modes are possible.

WORKED EXAMPLE

Five observations

Data $2, 3, 3, 7, 10$: mean $= 25/5 = 5$, median $= 3$, mode $= 3$. Outlier 10 pulls the mean more than the median.

For a discrete frequency table, locate the middle cumulative-frequency positions. For continuous grouped classes of common width h, let l be the lower boundary of the median class, c_f cumulative frequency before it, and f its frequency. Linear interpolation gives $\text{Median} = l + \frac{N/2 - c_f}{f}h$. This estimates a grouped distribution, not the unknowable exact raw median.

Mean deviation

Mean deviation about a reference a is the average absolute distance:

$$MD_a = \frac{\sum_i f_i|x_i - a|}{N}.$$

Often asked about the mean or median; specify the centre. Absolute values prevent positive/negative deviations cancelling.

COMMON TRAP

Use absolute values

$\sum f_i(x_i - \bar{x}) = 0$ by construction, so the signed mean deviation is not a spread measure. Use $|x_i - a|$.

KEY RESULT

Grouped continuous classes

Use class marks

$x_i = (\text{lower boundary} + \text{upper boundary})/2$

as representative observations for mean/variance estimates. Results are estimates because original within-class positions are unknown.

Population variance and standard deviation

For data x_i with frequencies f_i, mean $\bar{x}$ and $N = \sum f_i$:

$$\sigma^2 = \frac{\sum_i f_i(x_i - \bar{x})^2}{N} = \frac{\sum_i f_i x_i^2}{N} - \bar{x}^2, \qquad \sigma = \sqrt{\sigma^2}.$$

This syllabus convention treats the listed data as the full distribution. Sample variance using divisor $N - 1$ is a different inferential estimator; do not silently substitute it.

SHORT DERIVATION

Computational identity

Expand

$$\sum f_i(x_i - \bar{x})^2 = \sum f_i x_i^2 - 2\bar{x}\sum f_i x_i + N\bar{x}^2 = \sum f_i x_i^2 - N\bar{x}^2.$$

.

Shift and coding shortcuts

Adding constant c shifts mean and median by c, leaves variance/SD unchanged. Multiplying by a scales mean by a, variance by a^2, SD by $|a|$. For equal class width h, coded values $u_i = (x_i - A)/h$ give $\sigma_x = h\sigma_u$.

WORKED EXAMPLE

Frequency table

Values $1, 2, 3$ with frequencies $1, 2, 1$: $N = 4$, mean 2, mean square $= (1 + 8 + 9)/4 = 4.5$, variance $= 0.5$, SD $= \sqrt{0.5}$.

Grouped continuous data

For classes $[a_i, b_i)$ with frequency f_i, class mark $m_i = (a_i + b_i)/2$ approximates every value in the class. Use m_i in the frequency formulas for mean and variance. Median class is first cumulative frequency at least $N/2$; estimated median is $l + \frac{N/2 - c_f}{f_m}h$. For a unique modal class in equal-width classes of width h, lower boundary l, modal-class frequency f_1 and adjacent frequencies f_0, f_2, the standard interpolation estimate is $\text{Mode} = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2}h$. These interpolations assume a reasonably smooth distribution within/around the relevant classes.

WORKED EXAMPLE

Grouped table estimate

Classes $[0,10), [10,20), [20,30)$ have frequencies $4, 8, 4$ ($N = 16$). Midpoints $5, 15, 25$ give mean 15, variance 50, SD $\sqrt{50}$. The middle class has $l = 10, c_f = 4, f_m = 8$, so grouped median estimate is $10 + (8-4)10/8 = 15$. Its mode estimate is also 15 by symmetry ($f_0 = f_2 = 4$). Mean deviation about 15 is $(4 \cdot 10 + 8 \cdot 0 + 4 \cdot 10)/16 = 5$. All are class-based estimates, not exact raw-data statistics.

COMMON TRAP

Unequal class widths

For a histogram with unequal class widths, bar height is frequency density f_i/width_i, not raw frequency. The modal class is determined by density when comparing unequal-width histogram bars.

A workflow that prevents most errors

1. Define outcomes / variable / data convention.
2. State equally-likely, independence, replacement, or class-mark assumptions.
3. Choose count, conditional probability, Bayes, distribution, or descriptive statistic.
4. Compute exactly when possible; check range, units, and limiting cases.

WORKED EXAMPLE

Two cards without replacement

Probability both are aces: $(4/52)(3/51) = 1/221$. The second conditional probability is $3/51$, not $4/52$; equivalently $\binom{4}{2}/\binom{52}{2}$.

Useful identities

For independent X_i: $E[\sum X_i] = \sum E[X_i]$ always; $\text{Var}(\sum X_i) = \sum \text{Var}(X_i)$ under independence. For Bernoulli indicators, $X = \sum I_i$ yields binomial mean and variance immediately.

KEY RESULT

Range and sanity checks

Probability lies in $[0,1]$; variance is nonnegative; SD has original units; mean lies between min and max. A grouped estimate cannot recover exact within-class data.

Official coverage and source gap

JEE Advanced includes probability plus descriptive statistics (mean, median, mode, mean deviation, variance and standard deviation for grouped/ungrouped data, and mean/variance of a random variable). The repository's Probability module covers probability deeply but does not provide a dedicated descriptive-statistics treatment; these statistics sheets are authored to the syllabus and standard definitions, with the class-mark approximation stated.

COMMON TRAP

Not a sampling-statistics chapter

The variance formula here uses divisor N for a complete data set / probability distribution. The unbiased sample estimator with divisor $N-1$, confidence intervals, and hypothesis testing are outside this syllabus-oriented set.

Geometric Probability, Ballot, Birthday

colour key: navy = prose · teal = headings · yellow = key results · red = common traps

Continuous sample spaces

Uniformly chosen points in a region: probability is favourable area / total area, provided the joint distribution is uniform in the stated coordinates. Coordinate choice and sampling procedure matter.

WORKED EXAMPLE

Broken stick

Two independent uniform break points on a unit stick make three pieces. Triangle condition is that no piece is at least $1/2$. The region of valid break-point pairs has one quarter the total area, so probability $1/4$.

Bertrand ballot theorem

If $a > b$ votes are randomly ordered, the probability A is strictly ahead after every vote is $(a-b)/(a+b)$. Reflection of paths at their first diagonal contact counts the bad orderings. Strictly ahead excludes ties after the first vote.

Birthday collision

With d equally likely birthdays, independent and ignoring leap day, probability of no match among $n \le d$ people is $\prod_{j=0}^{n-1} (d-j)/d$; collision probability is one minus this. For $d = 365, n = 23$, about 0.5073.

COMMON TRAP

Model assumptions matter

Uniform birthdays, independence, and a 365-day year are approximations. The answer is conditional on this idealised model, not a statement about real populations.

Build state equations

For a process with finite memory, let p_s be the desired probability from state s. Condition on the next move: $p_s = \sum_t P(s \to t) p_t$, using boundary values for winning/losing states. For expectations, add one unit of cost per step: $E_s = 1 + \sum_t P(s \to t) E_t$.

WORKED EXAMPLE

Alternating first-head game

Fair coin; players alternate, first head wins. Let p be first player's win chance. She wins now with $1/2$ or both tosses fail with $1/4$ and state resets: $p = 1/2 + p/4$, hence $p = 2/3$.

Coupon collector

For n equally likely types, when j distinct types are collected the next-new-type chance is $(n-j)/n$. Summing geometric stage means gives

$$E[T] = n\left(1 + \tfrac{1}{2} + \cdots + \tfrac{1}{n}\right) = nH_n.$$

Memorylessness and runs

Geometric waiting time: $P(X > m+n \mid X > m) = P(X > n)$. There is no "due" effect. Expected tosses to obtain r consecutive heads obey a run-state recurrence; for a fair coin it is $2^{r+1} - 2$.

Gambler's ruin

Fortune starts at i units, absorbing barriers $0, N$, and each unit bet wins with probability p, loses with $q = 1-p$. Success probability solves $u_i = p u_{i+1} + q u_{i-1}$, $u_0 = 0, u_N = 1$:

$$u_i = \frac{1-(q/p)^i}{1-(q/p)^N} \quad (p \neq q), \qquad u_i = i/N \quad (p = q = 1/2).$$

INTUITION

Fair fortune is a martingale

Expected final fortune equals initial fortune i; final fortune is N with probability u_i and zero otherwise, so $Nu_i = i$.

Derangements by inclusion–exclusion

For n objects assigned uniformly to n positions, probability none fixed is

$$\sum_{k=0}^{n} \frac{(-1)^k}{k!},$$

so $D_n = n!\sum_{k=0}^{n}(-1)^k/k!$ and it tends to $1/e$. At $n=5$, $D_5 = 44$ and probability $11/30$.

WORKED EXAMPLE

Biased walk, exact value

Take $p = 3/5$, $q/p = 2/3$, barriers 0 and 10, start $i = 5$. Then $u_5 = [1-(2/3)^5]/[1-(2/3)^{10}] \approx 0.88364$. The edge makes success more likely than $1/2$; the boundary conditions still matter.

COMMON TRAP

Do not overgeneralize optional stopping

The elementary fair-ruin argument is safe for finite barriers and the process stopped at absorption. General martingale stopping claims require hypotheses; do not invoke an unrestricted optional-stopping theorem.

Derangement proof and a small check

Let A_i be the event that object i is fixed. Inclusion–exclusion gives $D_n = \sum_{j=0}^{n}(-1)^j\binom{n}{j}(n-j)! = n!\sum_{j=0}^{n}\frac{(-1)^j}{j!}$. For $n=3$, this is $6-9+2=2$; probability $1/3$. For $n=5$, $D_5 = 44$, probability $11/30$. The limit $1/e$ is approached with alternating partial-sum error bounded by the next reciprocal factorial.

Why averaging helps

For a random variable X, $E[X] = \sum_i E[I_i]$ when $X = \sum_i I_i$, regardless of dependence. This linearity powers coupon-collector stage sums and indicator proofs of expected collision counts. Variance does not share this unrestricted additivity.

What these extensions teach

The same ideas recur: complement for rare collisions, inclusion–exclusion for forbidden fixed points, state recursions for sequential games, and expectation decomposed into stages. These models add methods; they are not explicit JEE syllabus promises.